Planning-Guided Failure-Triggered Recovery via Imitation Learning for Dynamic Social Navigation

Anonymous Authors

Abstract—Classical navigation stacks are reliable in routine mapped environments but can fail in reciprocal interactions with moving people. End-to-end learned navigation can address such interactions, yet replacing a mature planner everywhere changes behavior even where no recovery is needed. We study a bounded alternative: an observable failure detector invokes a learned policy only during collision risk, freezing, oscillation, or deadlock. The policy chooses among 21 temporary subgoals and four interpretable modes, while a planning-derived mask rejects invalid actions and Nav2 executes each selected goal. A privileged short-horizon expert supplies imitation labels and DAgger covers learner-induced recovery states. During validation we found that simulator command acceptance was insufficient evidence that sensed geometry, localization, and evaluation truth agreed. The corrected fallback reads robot and pedestrian evaluation poses from Gazebo, separates rendered pedestrians from contact dynamics, and uses pose-derived localization. Across two fixed high-density crossing-flow seeds, DWB collided twice, while triggered DAgger physically reached the original goal twice and increased median minimum human-centre distance from 0.642 to 1.075 m. These executions establish a repeatable preliminary signal, not statistical superiority; a locked multi-scenario evaluation remains required.

I. Introduction

Classical navigation offers explicit maps, reusable planners, and predictable nominal behavior, but local optimization can freeze or oscillate in interactive bottlenecks. Learned policies can encode interaction patterns, but continuous end-to-end replacement expands the learned component's authority and makes failures harder to interpret.

We instead confine learning to the failure boundary. Figure 1 shows the hierarchy. A classical planner owns nominal PointGoal navigation. An observable detector and hysteretic state machine trigger a bounded recovery option. A compact policy chooses an interpretable temporary target or mode from a fixed action set, subject to one mask shared by data generation and deployment. Once progress resumes, the original task goal is reissued.

The current contribution is threefold: (i) a failure-triggered architecture that preserves classical nominal control; (ii) an automated privileged planning expert that labels masked temporary-goal choices; and (iii) an imitation pipeline using behavior cloning and DAgger. We do not claim a first recovery system or a formal safety guarantee. This evidence-linked draft reports the corrected simulator-validity result and does not reuse pre-verified-pose comparisons as performance evidence.

II. Related Work

Arena-Bench and Arena-Rosnav provide reproducible interfaces for comparing model-based and learned navigation in dynamic environments [1], [2], [3], [4]. All-in-One learns to select among navigation planners [5]; DR-MPC learns a residual around model-predictive control [6]. Both motivate hybrid navigation but differ from failure-only intervention.

Learned navigation recovery predates this work. Del Duchetto et al. learn failure detection and local recovery from human demonstrations [7]. Mohammad et al. proactively predict planning failure and search for a recovery state [8]. Recent failure-aware learning uses unsafe experience to shape value estimation while restricting policy learning to successful demonstrations [9]. Our narrower distinction is automatic recovery demonstrations from privileged rollout planning, temporary-subgoal decisions, and aggregation on learner-induced recovery states. DAgger is the standard mechanism for mitigating sequential distribution shift [10].

III. Method

A. Failure-Triggered Hierarchy

Let g be the original goal and let the classical stack produce command u_b. From observable history o_t, the detector returns scores for collision risk, freeze, oscillation, and deadlock. Recovery begins after the maximum score exceeds τ_{on} and exits only after it remains below $\tau_{\text{off}} < \tau_{\text{on}}$ while valid goal progress resumes. Cooldown, option duration, and recovery-count limits prevent rapid switching.

B. Action Space and Mask

The action space contains temporary goals at radii $\{0.6, 1.0, 1.4\}$ m and bearings $\{-90, -60, -30, 0, 30, 60, 90\}^\circ$, plus WAIT, BACKUP, REPLAN, and CONTINUE. The mask rejects occupied, disconnected, LiDAR-occluded, rear-unobserved, or path-divergent motion. During latched collision risk it also blocks unsafe continuation and applies endpoint and swept-segment clearance. Masking precedes softmax, so invalid actions cannot be selected while a valid fallback remains.

An independent stopping layer uses

$$d_{\text{stop}} = \frac{v^2}{2a_{\text{brake}}} + v\ell + m, \tag{1}$$

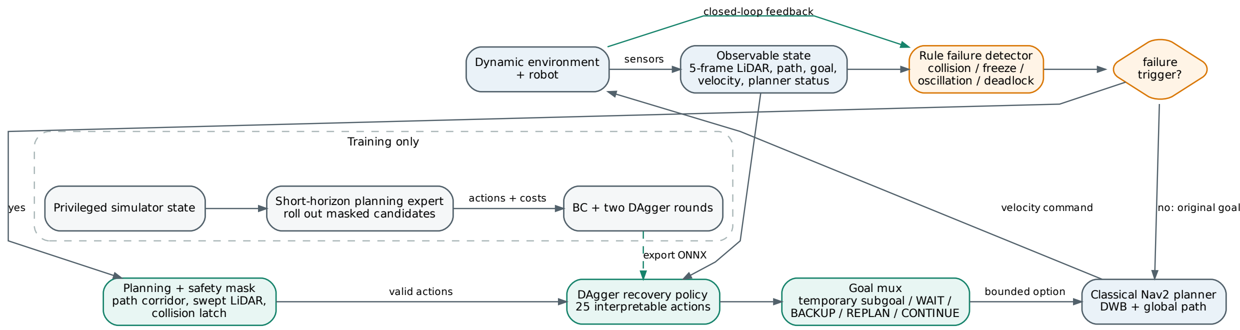

Fig. 1. Closed-loop architecture. Privileged state and expert rollouts are used only during training. At test time, the policy receives observable state and may only choose actions admitted by the planning and safety mask.

where ℓ is control latency and m an empirical margin. It may override learning but is not a formal collision-avoidance certificate.

C. Privileged Expert and Imitation

During training only, the expert receives the static grid and true pedestrian states. Each legal candidate is rolled out with differential-drive kinematics and predicted human motion. It minimizes collision, progress, social distance, path rejoin, length, smoothness, time, and action-switch costs; collision and masked actions receive hard penalties. The policy uses a one-dimensional CNN over five 180-beam scans and an MLP over observable navigation state. Behavior cloning minimizes expert cross entropy, and DAgger relabels policy-visited training states with the privileged expert.

IV. Experimental Setup

Experiments use ROS2 Humble, Nav2 DWB, and a pinned Arena Humble Gazebo profile. The installed profile lacks a complete HuNav dependency chain, so deterministic LiDAR-visible fallback pedestrians are disclosed. The robot receives a known static map and a planar scan downsampled to 180 beams. Train, validation, and test splits are scenario-separated.

Fallback pedestrians are non-static, gravity-disabled rendered cylinders without contact response; overlap is classified geometrically from Gazebo model poses. This prevents teleported proxy bodies from imparting non-human impulses to the Jackal while preserving GPU-LiDAR visibility. Gazebo's pose-derived odometry supplies the declared known-pose localization interface; a real mapped deployment would replace it with localization such as AMCL. An evaluation-only gate projects human centres into the scan and checks that near-human surfaces are visible at expected bearings. Human privileged data are never policy inputs. All valid outcomes are retained;

TABLE I
Verified-pose medium crossing-flow validation pair. Goal distance and human clearance use Gazebo model feedback; this is single-seed execution evidence only.

Method	Outcome	Time [s]	d_g^{phys} [m]	$d_{\min}$ [m]	Recovery
Classical DWB	COLLISION	29.5	14.074	0.665	0
Triggered DAgger	GOAL REACHED	98.4	0.299	1.203	117

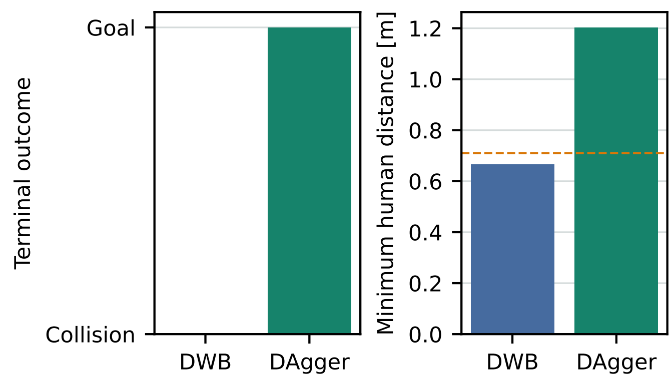

Fig. 2. First verified-pose pair. DWB collides; learned recovery reaches the physical goal while maintaining greater human clearance.

simulator/reset failures are reported separately. The reported pilot pairs use validation scenarios and are not a final test evaluation.

V. Corrected Validation Check

The static-proxy implementation produced an impossible observation: privileged evaluation placed a pedestrian centre 0.697 m from the robot while the LiDAR sector at that bearing remained 1.473 m away. A non-static kinematic-link revision still allowed model pose requests to diverge from rendered geometry. A later dynamic collision-body revision pushed the Jackal, and wheel-integrated odometry declared arrival while the physical robot remained 1.06 m from the goal. Consequently,

TABLE II
High-density crossing-flow validation pilot on two fixed seeds. Values are descriptive medians; $n = 2$ is insufficient for significance testing.

Method	Goal	Collision	Time [s]	$d_{\min}$ [m]	Recovery
Classical DWB	0/2	2/2	29.8	0.642	0.0
Triggered DAgger	2/2	0/2	116.9	1.075	258.5

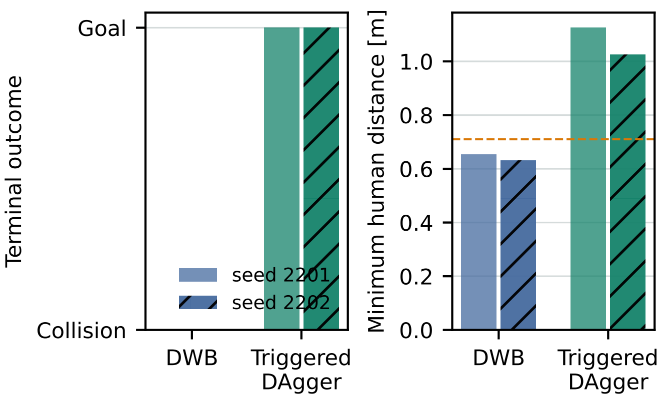

Fig. 3. Two-seed high-density pilot under one frozen code revision. Both DWB runs collided and both triggered DAgger runs physically reached the goal. The dashed line is the 0.71 m geometric robot–human collision threshold.

every pre-verified-pose comparison is retained only for debugging. The final path reads evaluation poses from Gazebo's dynamic-pose topic, uses rendered contactless pedestrians, and checks physical goal distance independently of localized distance.

Table I and Fig. 2 summarize the first verified-pose pair from commit 2164e08. DWB collided at 29.47 s with 0.665 m minimum centre distance. The learned hierarchy physically reached the goal at 98.43 s after 117 non-CONTINUE recovery samples, with 1.203 m minimum centre distance and 0.299 m final physical goal error. Its sensor gate passed 30/30 near-human frames; Base passed 26/26. Maximum localization–physical position disagreement was 0.128 m.

The subsequently frozen high-density pilot in Table II and Fig. 3 uses two independent scenario seeds under commit 6cf9535. DWB collided after 29.50 and 30.00 s. Triggered DAgger reached the physical goal after 114.92 and 118.91 s, ending 0.294 and 0.292 m from the goal. The learned runs used 254 and 263 non-CONTINUE samples, including WAIT, BACKUP, and temporary subgoals; thus the outcome was not obtained by disabling intervention. All four sensor gates passed at 100% visibility over 26, 180, 31, and 200 near-human frames. The pilot suggests a safety–time trade-off: successful recovery required about four times the baseline's collision-terminal duration. With $n = 2$, no hypothesis test or general performance claim is justified.

VI. Limitations

The method assumes a known two-dimensional map and planar LiDAR. Its privileged expert depends on simulator state and constant-velocity pedestrian prediction. The finite discrete action set cannot express arbitrary maneuvers. The Gazebo fallback differs from real crowds and exhibits scheduling artifacts. Pose-derived simulation localization is more accurate than a real localization stack. There is no formal safety guarantee, hardware study, completed learned detector, or PPO refinement. The learned policy is conservative and substantially increases navigation time in the reported high-density pilot. Corrected multi-family training data and the final locked multi-scenario test remain incomplete.

VII. Conclusion

Failure-triggered temporary-goal recovery can be integrated with a classical ROS2 navigation stack and trained from an automated privileged planner without granting the learned policy continuous velocity authority. The simulator-validity gate exposed and corrected hidden proxy and odometry-truth errors. Under the verified-pose implementation, the learned hierarchy converted DWB collisions into physical goal reaches on two fixed high-density crossing-flow seeds, at the cost of much longer navigation time. This is encouraging execution evidence rather than a general claim. The required next steps are broader corrected-data training and a locked paired multi-scenario evaluation.

References

[1] L. Kästner, T. Bhuiyan, T. A. Le, E. Treis, J. Cox, B. Meinardus, J. Kmiecik, R. Carstens, D. Pichel, B. Fatloun, N. Khorsandi, and J. Lambrecht, "Arena-bench: A benchmarking suite for obstacle avoidance approaches in highly dynamic environments," arXiv preprint arXiv:2206.05728, 2022.

[2] L. Kästner, R. Carstens, H. Zeng, J. Kmiecik, T. Bhuiyan, N. Khorsandi, V. Shcherbyna, and J. Lambrecht, "Arena-rosnav 2.0: A development and benchmarking platform for robot navigation in highly dynamic environments," arXiv preprint arXiv:2302.10023, 2023.

[3] L. Kästner, V. Shcherbyna, H. Zeng, T. A. Le, M. H.-K. Schreff, H. Osmaev, N. T. Tran, D. Diaz, J. Golebiowski, H. Soh, and J. Lambrecht, "Arena 3.0: Advancing social navigation in collaborative and highly dynamic environments," arXiv preprint arXiv:2406.00837, 2024.

[4] V. Shcherbyna, L. Kästner, D. Diaz, H. G. Nguyen, M. H.-K. Schreff, T. Lenz, J. Kreutz, A. Martban, H. Zeng, and H. Soh, "Arena 4.0: A comprehensive ros2 development and benchmarking platform for human-centric navigation using generative-model-based environment generation," arXiv preprint arXiv:2409.12471, 2024.

[5] L. Kästner, J. Cox, T. Bhuiyan, and J. Lambrecht, "All-in-one: A drl-based control switch combining state-of-the-art navigation planners," arXiv preprint arXiv:2109.11636, 2021.

[6] J. R. Han, H. Thomas, J. Zhang, N. Rhinehart, and T. D. Barfoot, "Dr-mpc: Deep residual model predictive control for real-world social navigation," arXiv preprint arXiv:2410.10646, 2024.

[7] F. D. Duchetto, A. Kucukyilmaz, L. Iocchi, and M. Hanheide, "Do not make the same mistakes again and again: Learning local recovery policies for navigation from human demonstrations," IEEE Robotics and Automation Letters, vol. 3, no. 4, pp. 4084–4091, 2018.

[8] N. Mohammad, J. Higgins, and N. Bezzo, “A gp-based robust motion planning framework for agile autonomous robot navigation and recovery in unknown environments,” arXiv preprint arXiv:2402.01617, 2024.

[9] X. Wang, S. Cheng, S. Wang, X. Zhang, D. Zhang, and W. Zhang, “Learning from demonstration with failure awareness for safe robot navigation,” arXiv preprint arXiv:2604.23360, 2026.

[10] S. Ross, G. J. Gordon, and J. A. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” in Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics, vol. 15. PMLR, 2011, pp. 627–635.